

A Comparison of Symbolic and Neuro-Symbolic Strategies for LTLf Runtime Monitoring

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Abstract

TODO

1 Introduction

Linear Temporal Logic (LTL) [16] is a logic formalism for expressing temporal properties of reactive systems. It extends traditional propositional logic with modal operators, allowing the specification of rules that must hold *through time*, and it is interpreted over traces of discrete events. Variants of LTL interpreted over finite traces have been proposed for modeling finite, but length-unbounded, process executions (see Sec. 2.1), making it suitable for finite-horizon problems that often arise in the context of AI and BPM. Temporal logics have provided one of the most popular tool in Formal Methods for verification of hardware or software system behaviors. In particular, model checking is the problem of verifying whether a model of a system meets a given specification. Despite the importance and the progress made in model checking, it is not always possible to check the model against the specification. This is due to the inherent complexity of the system or the fact that the model is only partially correct or not known. In such cases, one can resort to simpler verification approaches, such as runtime monitoring, where the system’s execution is continuously observed on-the-fly and any permanent violation or fulfillment of the specified property is immediately detected.

Standard symbolic techniques for both model checking and runtime monitoring have achieved great success, but they also suffer for some inherent limitation. Among the others there is the difficulty in dealing with the increasing growth of big data and the lack of flexibility in dealing with highly unpredictable environments, say e.g. environments with noisy sensors or probabilistic uncertainties, and adapting the specification when soft violations could happen. Conversely, pure deep learning architectures, such as Recurrent Neural Networks (RNNs) and Transformers, excel at modeling high-dimensional, noisy data distributions, but they likely generate outputs that violate critical safety constraints, as they lack a mechanism to adhere to formal logics. This naturally leads to the use of hybrid approaches, which can possibly overcome some of the mentioned limitations.

In this paper, we will compare different techniques for runtime monitoring, from purely symbolic methods to the state-of-the-art neuro-symbolic architectures. We start revisiting the neural-symbolic approach to runtime monitoring, originally proposed by [14, 15] through the RuleRunner system [13]. The authors claimed that the resulting neural monitors could exploit GPU parallelism for efficient verification, outperforming purely symbolic mechanisms. Since then, some significant developments have reshaped the landscape. First, LTL_f [4] has become the standard formalism for

temporal reasoning over finite traces in AI, bringing with it mature tools for compiling formulas into DFAs. Then, differentiable automata representations such as DeepDFA [17] have emerged, enabling the integration of temporal logic knowledge into neural architectures through gradient-based learning. **Possibly include other NeSy approaches.** We present a systematic comparison of three paradigms for LTL_f runtime monitoring: (i) purely symbolic DFA-based monitoring, (ii) rule-based neural-symbolic monitoring in the RuleRunner tradition, and (iii) automata-based neural-symbolic monitoring using DeepDFA.

TODO: Mention briefly the experimental results obtained.

2 Background

2.1 An Historical Overview of Temporal Logics over Finite Traces

Studying LTL over finite traces gained significant attention [8, 11]. The original RuleRunner system [13] refers to FLTL as a finite-trace variant of LTL. As already mentioned, in the last decade LTL_f has established as the standard language for modeling finite traces of executions. Notably, FLTL and LTL_f share essentially the same syntax and semantics, offering two next operators, with a strong next operator X which is false at the last position if no successor exists, and a weak next operator W which is true at the last position (see Sec. 2.2 for details). FLTL makes explicit use of an *END* symbol for signaling the last element of the trace, but this can be easily expressed with the weak next operator by $END \equiv W\text{false}$. In this paper, we use LTL_f as the specification language for all three monitoring paradigms we compare.

TODO: Mention other finite-trace variants.

2.2 Syntax and Semantics

Given a finite set $\mathcal{P}$ of atomic propositions, an LTL_f formula φ has the form:

$$\varphi ::= \top \mid \perp \mid a \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid X\varphi \mid \varphi_1 U \varphi_2,$$

where $a \in \mathcal{P}$. We use $\top$ and $\perp$ to denote *True* and *False*, respectively. X (Strong Next) and U (Until) are temporal operators. Other temporal operators can be derived as follows: W (Weak Next) is defined as $W\varphi \equiv \neg X\neg\varphi$, R (Release) is defined as $\varphi_1 R \varphi_2 \equiv \neg(\neg\varphi_1 U \neg\varphi_2)$, G (globally) is defined as $G\varphi \equiv \perp R \varphi$, and F (eventually) is defined as $F\varphi \equiv \top U \varphi$.

A trace $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_T)$ is a sequence of propositional assignments to the propositions in $\mathcal{P}$, where $\sigma_t \in 2^{\mathcal{P}}$ is the set of all and only propositions that are true at instant t . Additionally, $|\sigma| = T$ denotes the length of the trace. Note that, if the propositional symbols in $\mathcal{P}$ are all *mutually exclusive*, i.e., the domain produces exactly one symbol true at each step, then we have $\sigma_t \in \mathcal{P}$. Let $last = |\sigma| - 1$. We inductively define when an LTL_f formula φ is true at an instant i (for $0 \leq i \leq last$), written $\sigma, i \models \varphi$, as follows:

- $\sigma, i \models a$ iff $a \in \sigma_i$, for $a \in \mathcal{P}$;
- $\sigma, i \models \neg\varphi$ iff $\sigma, i \not\models \varphi$;
- $\sigma, i \models \varphi_1 \wedge \varphi_2$ iff $\sigma, i \models \varphi_1$ and $\sigma, i \models \varphi_2$;
- $\sigma, i \models X\varphi$ iff $i < last$ and $\sigma, i + 1 \models \varphi$;

- $\sigma, i \models \varphi_1 \mathcal{U} \varphi_2$ iff for some j such that $i \leq j \leq \text{last}$, we have $\sigma, j \models \varphi_2$, and for all k such that $i \leq k < j$, we have $\sigma, k \models \varphi_1$.

We say that σ satisfies ϕ , written $\sigma \models \varphi$, if $\sigma, 0 \models \varphi$. A formula φ is satisfiable if it is true wrt some trace σ , and is valid, if it is true wrt every trace σ .

In the following, we will use $\text{sub}(\varphi)$ to denote the set of proper subformulas of φ , and we will refer to it as the syntactic closure of φ . Given a formula φ and a propositional assignment σ_t at time t , we are also interested in computing the residual formula, i.e the formula that has to be satisfied at time $t + 1$. As explained in details in Sec. ??, a residual formula can be computed using formula progression $\square$, which returns a positive Boolean formula on $\text{sub}(\varphi) \cup \text{sub}(F\top) \cup \text{sub}(G\perp)$. We will use $\text{cl}(\varphi)$ to denote the set of residual formulas, and we will refer to it as the residual closure of φ .

2.3 Symbolic Monitoring

Deterministic Finite Automata. Any LTL_f formula φ can be translated into a Deterministic Finite Automaton (DFA) [4] $A_\varphi = (2^P, Q, q_0, \delta, F)$, where 2^P is the automaton alphabet, Q is the finite set of states, $q_0 \in Q$ is the initial state, $\delta : Q \times 2^P \rightarrow Q$ is the transition function, and $F \subseteq Q$ is the set of final states. We also define the extended transition function over traces $\delta^* : Q \times (2^P)^* \rightarrow Q$ recursively as:

$$\begin{aligned} \delta^*(q, \epsilon) &= q \\ \delta^*(q, \sigma + \mathbf{x}) &= \delta^*(\delta(q, \sigma), \mathbf{x}), \end{aligned} \tag{1}$$

where $\sigma \in 2^P$ is a symbol and $\mathbf{x} \in (2^P)^*$ is a trace. The automaton accepts the trace σ iff $\delta^*(q_0, \sigma) \in F$, and in that case we say that σ belongs to the language of the automaton, denoted as $\mathcal{L}(A_\varphi)$. We have that φ and A_φ are equivalent because, for any trace $\sigma \in (2^P)^*$:

$$\sigma \in \mathcal{L}(A_\varphi) \iff \sigma \models \varphi. \tag{2}$$

We can rely on the use of DFAs for several reasoning tasks, like model checking, planning and synthesis. In particular, monitoring formula φ becomes a deterministic walk over the automaton A_φ : the monitor keeps a single current state q , initialized to q_0 , and on observing the time point $\sigma_t \in 2^P$ it advances $q \leftarrow \delta(q, \sigma_t)$. Because the automaton is fixed and deterministic, each step costs a single transition lookup, independent of the trace seen so far and of the size of φ . In the following, DFA-based monitoring will serve as the exact baseline against which the neuro-symbolic paradigms will be measured.

Verdicts and early termination. Online monitoring requires a three-valued runtime semantics [2]: at each step the monitor emits SATISFY, VIOLATE, or UNDECIDED, the last meaning that the observed prefix can still be extended into both an accepting and a non-accepting continuation. A definite verdict can be issued as soon as the outcome is settled regardless of the future, which corresponds to two structural properties of the current state, both precomputed once by backward reachability over A_φ . A state is a *trap* if no accepting state is reachable from it: reaching a trap means φ is permanently violated, so the monitor halts with VIOLATE. Dually, a state is an *accepting sink* if every state reachable from it (including itself) is accepting: the monitor halts with SATISFY. With trap and sink sets materialized at compile time, the per-step verdict reduces to two set-membership tests on top of the transition lookup. As long as the current state is neither, the

verdict is UNDECIDED and monitoring continues. Some specifications never reach a trap or a sink, so STEP returns UNDECIDED for the entire trace. An example is the response pattern $G(a \rightarrow Fb)$, which can always be satisfied by a later b and always be rescued from violation, hence has neither an accepting sink nor a trap. For these the verdict is only binary at the end of the trace, and the monitor reports a final verdict by testing acceptance of the state reached, i.e. SATISFY iff $\delta^*(q_0, \sigma) \in F$ and VIOLATE otherwise.

3 Rule-based Neuro-Symbolic Monitoring: RuleRunner

3.1 Architecture

A RuleRunner system is a tuple $\langle S, R_E, R_R \rangle$, where S is the state of the system, R_E is the set of *evaluation rules*, and R_R is the set of *reactivation rules*. In details:

- The state S contains observations, active rules, and truth evaluations. $R[\phi]$ denotes an active rule, while $[\phi]V$ denotes a 3-values truth evaluation, with $V \in \{T, F, ?\}$.
- Evaluation rules are used to compute the truth value of a formula ϕ under verification, given the truth values of its direct subformulas ψ_i . They have the form $R[\phi], [\psi_1]V, \dots, [\psi_n]V \rightarrow [\phi]V$. For instance, $R[a \vee b], [a]T \rightarrow [a \vee b]T$, because the disjunction is evaluated to *true* if one of the two disjuncts is true.
- Reactivation rules are used when a formula ϕ is evaluated to undecided, i.e. $[\phi]?$, in which case that formula and its subformulas are scheduled to be monitored again at the next time point of the trace. They have the form $[\phi]? \rightarrow R[\phi], R[\psi_1], \dots, R[\psi_n]$. For instance, $[\Diamond b]? \rightarrow R[b], R[\Diamond b]$, because if $\Diamond b$ is undecided, we need to re-evaluate b and $\Diamond b$ at the next step.

Both active rules and truth evaluations can be enriched by a *qualifier*, such as B (binary), L (left) and R (right), which provides additional information recording which operands within a formula/subformula are still being watched. For instance, $R[a \wedge \Diamond b], [a]T \rightarrow [a \wedge \Diamond b]?R$, because if the left conjunct a is true, it is still needed to check the truth value right conjunct $\Diamond b$ to assign a truth value to the conjunction.

Verification of a trace against a formula proceeds iteratively. At the beginning of each iteration (corresponding to the monitoring of a time point), the state S contains a set of rule names corresponding to the subformulas to be checked at that time point. First, the procedure includes observations of the current time point in the state S and fires the evaluation rules repeatedly incrementing the state itself until it reaches a fixpoint. In practice, this means computing the truth values of all subformulas $sub(\phi)$ of ϕ in a bottom-up way, from simple atoms to ϕ itself (see Figure ...). If the root evaluates to true (resp. false), the monitor halts with verdict SATISFY (resp. VIOLATE). Otherwise, the reactivation rules are fired, and the result constitute the state of the next iteration. Note that in this case the state is not expanded, but replaced by the result of reactivation rules. This is used to flush all the previous truth evaluation, which are to be computed from scratch at the new time point. The procedure terminates with a decided verdict because of special END-triggered evaluation rules, that returns either true or false if we consumed the whole trace. A RuleRunner system can be translated in a logic program, which in turn can be encoded in a Recurrent Neural Network with a single hidden layer, s.t. each evaluation and reactivation rule

is a mapping from the input layer to the output one. For a detailed description of RuleRunner and verification algorithm see [14, 13].

The rule system underlies two behaviorally-equivalent neural encodings, both of which we evaluate in Section 6. In the *flat* encoding [14], the whole rule set is compiled into a single recurrent network following the standard CILP translation []: one hidden unit per rule, weights realizing each Horn clause, and the fixpoint of the evaluation phase recovered by iterating the forward pass until the fixpoint is met. In the *structured* encoding [15], the same rules are instead grouped as one CILP subnetwork per parse-tree node, wired according to the syntax of φ . Operationally, we execute one bottom-up sweep of the evaluation modules and then fire every reactivation module independently, taking the union of their outputs as the recurrent state of the next cell. This is functionally equivalent to horizontally composing the modules into the single recurrent network of Figure 5 in [15]. The modular form is the substrate for *local learning*: because parameters are attached to syntactically-local subnetworks, adaptation can be confined to the subtree responsible for a misclassification rather than diffused across a monolithic network.

3.2 The Nested-Temporal Limitation

RuleRunner’s defining design choice is to maintain *a single shared state, with exactly one truth register per syntactic subformula*. This is what gives it a constant, compile-time rule count, a fixed-size network, and a non-branching state, and it is well suited to many flat temporal specifications. The problem is that the architecture *assumes each subformula has exactly one live truth value per time point*. This assumption is violated precisely when a temporal operator is nested inside the operand of another operator that may reinstall that operand across successive time points: the outer operator spawns a fresh instance of the operand while a prior instance of the *same* subformula is still resolving, and the two distinct live obligations are conflated onto the one shared register.

Consider the two formulas $\varphi_g = F(a \wedge b)$ and $\varphi_b = F(a \wedge Xb)$, which differ only by a single next operator, monitored over the same trace $\sigma = (\{a\}, \emptyset, \{b\})$. Neither formula is satisfied: φ_g would require a time point where a and b hold *together* (but a occurs only at $t = 0$ and b only at $t = 2$), and φ_b would require a time point t with $a \in \sigma_t$ and $b \in \sigma_{t+1}$ (but the only a is at $t = 0$, and is not followed by a b at $t = 1$). The correct verdict in both cases is therefore VIOLATE, but RuleRunner can provide the correct answer only for the first formula.

First, consider φ_g . The run is:

t	σ_t	evaluation	reactivation
0	$\{a\}$	$[a]T, [b]F \rightarrow [a \wedge b]F$ $[a \wedge b]F \rightarrow [F(a \wedge b)]?$	$\rightarrow R[F(a \wedge b)], R[a], R[b], R[a \wedge b]$
1	$\emptyset$	$[a]F, [b]F \rightarrow [a \wedge b]F$ $[a \wedge b]F \rightarrow [F(a \wedge b)]?$	$\rightarrow R[F(a \wedge b)], R[a], R[b], R[a \wedge b]$
2	$\{b\}$	$[a]F, [b]T \rightarrow [a \wedge b]F$ $[a \wedge b]F \rightarrow [F(a \wedge b)]?$ $[F(a \wedge b)]? \rightarrow_{\text{END}} [F(a \wedge b)]F$	$\rightarrow \text{VIOLATE}$

Now, consider φ_b . In this case, the run diverges:

t	σ_t	evaluation	reactivation
0	$\{a\}$	$R[Xb] \rightarrow [Xb]?$	$\rightarrow R[Xb]M, R[b]$
		$[a]T, [Xb]? \rightarrow [a \wedge Xb]?R$	$\rightarrow R[a \wedge Xb]R$
		$[a \wedge Xb]?R \rightarrow [F(a \wedge Xb)]?$	$\rightarrow R[F(a \wedge Xb)], R[a], R[Xb], R[a \wedge Xb]$
1	$\emptyset$	$R[Xb] \rightarrow [Xb]?$	$\rightarrow R[Xb]M, R[b]$
		$R[Xb]M, [b]F \rightarrow [Xb]F$	
		$R[a \wedge Xb]R, [Xb]? \rightarrow [a \wedge Xb]?R$	$\rightarrow R[a \wedge Xb]R$
		$R[a \wedge Xb]R, [Xb]T \rightarrow [a \wedge Xb]T$	
		$R[a \wedge Xb], [a]F \rightarrow [a \wedge Xb]F$	
2	$\{b\}$	$[a \wedge Xb]?R \rightarrow [F(a \wedge Xb)]?$	$\rightarrow R[F(a \wedge Xb)], R[a], R[Xb], R[a \wedge Xb]$
		$R[Xb] \rightarrow [Xb]?$	
		$R[Xb]M, [b]T \rightarrow [Xb]T$	
		$R[a \wedge Xb]R, [Xb]T \rightarrow [a \wedge Xb]T$	
		$R[a \wedge Xb], [a]F \rightarrow [a \wedge Xb]F$	
		$[a \wedge Xb]T \rightarrow [F(a \wedge Xb)]T$	$\rightarrow \text{SATISFY (wrong)}$

The divergence stems from RuleRunner’s single shared state, which keeps exactly one truth register per syntactic subformula. The decisive step is the F reactivation. At every time point, an F that remains undecided reinstalls a fresh $R[Xb]$ that will defer again, while the previous time point’s deferral is still resolving via $R[Xb]M$. As mentioned previously, the state is a single flat set of truth evaluations and rule names with no position index. A reinstalling operator can place, into that one set, a freshly-deferred instance of a subformula while a prior instance of the same subformula is still resolving. The two instances are indistinguishable, and the next evaluation pass conflates them, possibly holding contradictory values for one subformula at once. In other words, the reactivation phase does not preserve the single-instance-per-subformula invariant across the whole formula.

This is a structural consequence of the one-literal-per-subformula encoding, and does not depend on the encoding of RuleRunner or on the implementation artifacts. The conflation cannot be repaired at the output layer. Consider φ_b again, the two traces $\sigma_A = (\{a\}, \emptyset, \{b\})$ (ground truth VIOLATE) and $\sigma_B = (\emptyset, \{a\}, \{b\})$ (ground truth SATISFY) for φ_b . After the second time point both present the *identical* conflicted register $[Xb] = \{F, ?\}$, yet the correct resolutions are opposite: σ_A needs the F to win, σ_B needs the $?$ to win. Any monitor whose decision is a function of the shared register alone must answer identically on both, so no priority order over the register can be correct on both traces.

3.3 The Exact Boundary and a Bounded-Event Construction

The counterexample identifies the mechanism of failure, but not a syntactic boundary. Temporal nesting alone is neither necessary nor sufficient for a mismatch. For example, after the initialization and Next-mode rules are interpreted faithfully, $X(Xa)$ is handled correctly: the two Next nodes are distinct subformulas, and the inner one is activated only after the outer one advances. Conversely,

$$(Xa) \wedge X(Xa)$$

shares the subformula Xa between two temporal offsets. On the two-cell trace $(\emptyset, \{a\})$, the original RuleRunner returns SATISFY, whereas the correct verdict is VIOLATE. Operator identities can also make apparently dangerous nesting harmless. Moreover, the equivalent rewriting $X(a \wedge Xa)$ is certified correct, although $(Xa) \wedge X(Xa)$ is not: the former activates Xa once, whereas the latter

reaches its deduplicated register at two offsets. Hence an exact presentation must account jointly for global syntactic sharing and the semantic effect of each collision; this amounts to the reachability/equivalence analysis performed by the certifier, rather than a clean compositional grammar.

Exact semantic class. Let RR_φ be the deterministic finite-state monitor induced by the original RuleRunner rules, and let A_φ be a canonical DFA for φ . We define

$$\mathcal{C}_{RR} = \{\varphi \mid L(RR_\varphi) = L(A_\varphi)\}.$$

This is the maximal semantic class on which the original construction is correct. Membership is decidable exactly, rather than estimated from sampled formulas or traces. Both monitors have finite state, so we explore their synchronous product over $2^{\mathcal{P}_\varphi}$. A reachable product state whose two components disagree on final acceptance yields a counterexample; breadth-first exploration yields a shortest one. The same product also checks whether every definite online RuleRunner verdict agrees with the DFA’s trap or accepting-sink label. We call this latter property *prefix soundness*; it is weaker than requiring identical verdict timing, because RuleRunner may remain undecided after the DFA can already decide.

This certifier gives an exact per-formula boundary, but it does not explain how the construction behaves on a recognizable family of specifications.

A sufficient syntactic fragment. Let π range over propositional formulas and let $\mathcal{F}_{\text{flat}}$ be the Boolean closure of temporal clauses with propositional operands:

$$\begin{aligned} \pi &::= \top \mid \perp \mid p \mid \neg p \mid \pi \wedge \pi \mid \pi \vee \pi \mid \pi \rightarrow \pi, \\ \theta &::= \pi \mid X\pi \mid W\pi \mid F\pi \mid G\pi \mid \pi U \pi \mid \pi R \pi, \\ \varphi_{\text{flat}} &::= \theta \mid \varphi_{\text{flat}} \wedge \varphi_{\text{flat}} \mid \varphi_{\text{flat}} \vee \varphi_{\text{flat}} \mid \varphi_{\text{flat}} \rightarrow \varphi_{\text{flat}}, \end{aligned}$$

where $p \in \mathcal{P}$. Thus $(Fa) \rightarrow G(b \vee c)$ belongs to the fragment, whereas $F(a \wedge Xb)$ and $X(Xa)$ do not. The restriction is on temporal scope, not on the surrounding Boolean structure. In particular, this is a readable sufficient fragment rather than an attempt to reproduce the identity-sensitive semantic class $\mathcal{C}_{RR}$: the latter also contains formulas such as $X(Xa)$ and $F(Fa)$.

Theorem 1 (Flat-fragment correctness). *For every $\varphi \in \mathcal{F}_{\text{flat}}$, the original RuleRunner is language-equivalent to A_φ and is prefix-sound.*

Proof. Consider first one temporal clause θ . Every proper subformula below its temporal operator is propositional and therefore receives a definite Boolean value in the cell in which it is activated. Reactivating F , G , U , or R can consequently reinstall only propositional operand trees; no pending temporal instance is installed a second time. Similarly, X and W install a propositional operand exactly at the successor cell. The only temporal state carried between cells is therefore the state of θ itself, for which the RuleRunner modes and end-of-trace tables implement the corresponding finite-trace semantics.

Boolean rules do not reinstall their descendants. They only retain the mode recording which already-active operands remain relevant. Consequently, composing temporal clauses with the Boolean grammar above cannot create a second temporal instance. If an equal clause occurs more than once, all its occurrences were activated at the same trace position and denote the same obligation, so syntactic deduplication is harmless. Induction over cells, with a structural induction over the Boolean layer in each cell, now preserves the single-instance invariant and the intended

three-valued evaluation of every active formula. At the finite boundary the operator-specific end rules give $\sigma \models \varphi$ exactly. A definite value produced before the boundary is respectively true or false for every continuation by the same invariant, proving prefix soundness. $\square$

The inclusion $\mathcal{F}_{\text{flat}} \subseteq \mathcal{C}_{RR}$ can be strict, and membership in $\mathcal{C}_{RR}$ remains the exact test outside the fragment. We next give a second syntactic class that strictly enlarges $\mathcal{F}_{\text{flat}}$ while preserving a small, static RuleRunner network.

Bounded event formulas. Consider the syntactically bounded fragment

$$\beta ::= \top \mid \perp \mid p \mid \neg\beta \mid \beta \wedge \beta \mid \beta \vee \beta \mid \beta \rightarrow \beta \mid X\beta \mid W\beta.$$

Its future horizon is defined by

$$\begin{aligned} h(p) &= 0, & h(\top) = h(\perp) &= 0, \\ h(\neg\beta) &= h(\beta), & h(\beta_1 \circ \beta_2) &= \max(h(\beta_1), h(\beta_2)), \\ h(X\beta) &= 1 + h(\beta), & h(W\beta) &= 1 + h(\beta), \end{aligned}$$

where $\circ \in \{\wedge, \vee, \rightarrow\}$. Standard unbounded F , G , U , and R have infinite syntactic horizon. Semantic normalization may reveal special bounded cases containing these operators, but we keep that question separate from this transparent sufficient test.

Let $\beta_1, \dots, \beta_m$ be the maximal non-atomic bounded subformulas of φ . Replace each distinct β_i with a fresh event proposition e_i , obtaining a *RuleRunner skeleton* $\alpha = \text{sk}(\varphi)$. This gives the checkable bounded-event fragment

$$\mathcal{F}_{\text{BE}} = \{\varphi \mid \text{sk}(\varphi) \in \mathcal{F}_{\text{flat}}\}.$$

This two-level definition—a bounded grammar followed by a syntactic skeleton test—mirrors the construction more faithfully than a single grammar: bounded islands are evaluated by the front end, while only their skeleton is evaluated by the original RuleRunner. For example,

formula	bounded event	skeleton
$G(Xa)$	$e \equiv Xa$	Ge
$F(a \wedge Xb)$	$e \equiv a \wedge Xb$	Fe
$G(a \rightarrow Xb)$	$e \equiv a \rightarrow Xb$	Ge
$a U (b \wedge Xc)$	$e \equiv b \wedge Xc$	$a U e.$

By contrast, eventization does not change $G(a \rightarrow Fb)$, because Fb has no finite horizon; its skeleton is rejected by the exact certifier.

Let $H = \max_i h(\beta_i)$. For each non-empty trace $\sigma = \sigma_0 \cdots \sigma_{T-1}$, define a derived trace $D_\varphi(\sigma)$ of the same length. Its cell at position t retains the original propositions required by α and contains e_i exactly when

$$\sigma, t \models \beta_i.$$

A streaming transducer implements this definition with a fixed window. It stores the previous H observations; together with the current observation, these form the $H + 1$ cells needed to emit the oldest derived cell. At end-of-trace it flushes the remaining suffixes, interpreting strong Next as false and weak Next as true when no successor exists. The empty trace has no position at which an event can be emitted, so its truth value is computed directly from the empty-trace semantics of φ .

Theorem 2 (Bounded-event composition). *Let α be the eventized skeleton of an LTL_f formula φ . If the exact product certifies that RR_α is language-equivalent to A_α , then for every finite trace σ the composition of the bounded-event transducer and RR_α returns SATISFY iff $\sigma \models \varphi$. If RR_α is also prefix-sound, every verdict returned before end-of-trace is sound.*

Proof. For a non-empty trace, the definition of each e_i and structural induction on φ give the substitution equivalence

$$\sigma, 0 \models \varphi \iff D_\varphi(\sigma), 0 \models \alpha.$$

The fixed-window transducer emits exactly $D_\varphi(\sigma)$: a position at least H cells behind the input is mature, while the final flush supplies the finite boundary for the remaining positions. Language equivalence of RR_α therefore gives the correct final verdict. The empty trace follows from the separately computed boundary value.

For an early verdict, the emitted cells form a prefix of the derived trace. Prefix soundness states that the verdict is permanent under every continuation of that derived prefix. The continuations realizable by the bounded-event transducer form a subset of those continuations, so the same verdict is sound for every continuation of the original trace. $\square$

Soundness is not timeliness. The theorem deliberately claims no latency bound. The derived *prefix* trails the input by at most H cells, but the *verdict* need not: it is exactly the inclusion used in the proof that is lossy in the other direction. Eventization forgets which derived words are realizable — in particular that the final position of any trace falsifies every strong-Next event — so RR_α reasons about continuations that no original trace produces and may miss a permanence that already holds. The sharpest witness is $G(Xa)$, which is unsatisfiable on every non-empty trace, so A_φ traps after one cell. Its skeleton is Ge , and the derived word in which e holds everywhere is consistent with Ge ; the skeleton therefore stays undecided for the whole trace and reports VIOLATE only at the boundary. The lag is the trace length, not H . Recovering the DFA’s timing consequently requires information about which continuations are realizable, which is exactly what the extrapolation of Corollary 2 supplies.

Corollary 1 (Syntactic enlargement). *For every $\varphi \in \mathcal{F}_{\text{BE}}$, the bounded-event RuleRunner returns the correct final verdict and every early definite verdict is sound. Moreover,*

$$\mathcal{F}_{\text{flat}} \subsetneq \mathcal{F}_{\text{BE}}.$$

Proof. If $\varphi \in \mathcal{F}_{\text{BE}}$, its skeleton belongs to $\mathcal{F}_{\text{flat}}$ and hence satisfies both hypotheses of Theorem 2 by Theorem 1. Eventizing a flat-fragment formula only replaces bounded subformulas by atoms and cannot introduce temporal nesting, so $\mathcal{F}_{\text{flat}} \subseteq \mathcal{F}_{\text{BE}}$. The inclusion is strict: $F(a \wedge Xb)$ is not flat, but its bounded island $a \wedge Xb$ is replaced by e , leaving the flat skeleton Fe . $\square$

For comparison with the implementation’s exact admission check, define

$$\mathcal{C}_{\text{BE}} = \{\varphi \mid RR_{\text{sk}(\varphi)} \text{ is language-equivalent to } A_{\text{sk}(\varphi)} \text{ and prefix-sound}\}.$$

Writing $\mathcal{C}_{\text{prog}}$ for the correctness class of the progression construction, the two readable fragments and the exact classes satisfy

$$\mathcal{F}_{\text{flat}} \subsetneq \mathcal{F}_{\text{BE}} \subseteq \mathcal{C}_{\text{BE}} \subsetneq \mathcal{C}_{\text{prog}} = \text{Form}(\text{LTL}_f).$$

The last inclusion is strict for the present bounded syntax: for example, $G(a \rightarrow Fb)$ has no non-atomic bounded island and its unchanged skeleton fails certification. The equality is established by the progression correctness theorem below.

Exact online extrapolation. The theorem above already gives exact final verdicts, but the delayed skeleton alone may discover a permanent verdict arbitrarily late, even when H is fixed. We recover exact three-valued timing from the complete pipeline state. Let z_t contain the RuleRunner carry after the mature derived prefix, its last evaluated cell, the at-most- H observations still in the pipeline, and the fill level. These components induce a finite deterministic transition system over the original alphabet. Mark z accepting when flushing its pipeline and then ending the trace yields SATISFY. Reverse reachability gives

$$\begin{aligned} z \text{ is an accepting sink} &\iff \text{no rejecting state is reachable from } z, \\ z \text{ is a trap} &\iff \text{no accepting state is reachable from } z. \end{aligned}$$

The corresponding state cubes are translated into a fixed CILP readout with one hidden unit per reachable composite state. This is a compiled state classifier rather than a compositional circuit; it labels the current state only and does not compute recurrence.

We report this as a priced option rather than as part of the construction, because on the natural family it is more expensive than the object it avoids. The composite state contains the window, so its graph grows roughly as $2^{|P| \cdot H}$: for $G(a \rightarrow (b \vee Xb \vee \dots \vee X^4b))$ it has 613 states, whereas A_φ has 6. The default configuration of our implementation is therefore the construction of Theorem 2 — a fixed static pipeline with no residual closure and no automaton — and exact online timing is enabled explicitly. Certification itself is precomputed offline for the same reason: running the product check inside compilation would charge the pipeline for a DFA construction it never uses at run time.

Corollary 2. *After every non-empty prefix, the extrapolated bounded-event monitor returns a three-valued label identical to that of the canonical DFA for the original formula.*

Proof. For every reachable z_t , the composition argument of Theorem 2, applied to every continuation of the prefix, shows that paths from z_t end in an accepting state exactly for the continuations satisfying the original formula. Hence absence of a reachable rejecting (respectively accepting) state is exactly the DFA accepting-sink (respectively trap) condition. $\square$

Neural realization and relation to prior work. The transducer is compiled into the same two neural organizations as RuleRunner. A recurrent CILP layer implements the observation shift register. For every available suffix length $1, \dots, H + 1$, the bounded formulas are unrolled into two-valued Horn rules over literals $[\psi, k]T$ and $[\psi, k]F$, where k is an offset in the window. The flat realization pools these rules and iterates them to a fixpoint; the structured realization owns one module per (ψ, k) and evaluates the modules in dependency order. Their event outputs feed, respectively, the flat and structured CILP realizations of the certified original-RuleRunner skeleton. The two networks therefore differ only in rule grouping, as do the two original RuleRunner implementations.

Finite temporal horizons, maximal bounded separation formulas, and their evaluation through a pipeline are due to Finkbeiner and Kuhtz [5]. Their construction supports bounded Until and uses strong/weak truncated-path contexts together with an extrapolation circuit to recover the Boolean value under their finite truncated-path semantics. We do not add their interval-indexed Until syntax: every fixed finite bound can be unrolled into the Boolean and Next fragment β , so native support would improve only the succinctness of the input and is immaterial to our construction. Our construction specializes the idea to the RuleRunner architecture: the unbounded prefix component

is the original RuleRunner and is admitted only after exact certification. Because our monitor interface instead uses permanent three-valued labels, we classify the reachable composite state as described above rather than equating truncated truth with a sink or trap. The result is a strict middle construction: it repairs bounded-future aliasing without enumerating the full residual closure, while progression remains necessary when the certified skeleton is unsafe. Exact online classification does enumerate the reachable composite states over the original observation alphabet; that cost is exponential in $|\mathcal{P}| \cdot H$ and is reported separately from the pipeline it labels.

3.4 Progression-Based Reactivation

The counterexample of Section 3.2 should not be read as a failure of the RuleRunner idea as a whole. It isolates a more precise problem: the original construction carries information across time using registers addressed only by the syntactic subformulas of the input formula. The local evaluation rules are not the source of the error. Given the values of the direct subformulas at one time point, the extended truth tables compute the intended three-valued result. What fails is the reactivation phase: when the same subformula is reinstalled from different temporal contexts, the distinct residual meanings of that subformula are written into the same register.

The repair we consider keeps the two ingredients that make RuleRunner distinctive: a rule-based evaluation/reactivation cycle, and the possibility of translating the resulting rule system into a recurrent CILP-style network. The change is in what is carried across time. Instead of carrying only truth registers for subformulas of the original input formula, the repaired monitor carries residual formulas obtained by progression [?, ?]. The monitor still uses formula registers, and these registers are still associated with syntactic formulas; however, the relevant formulas are now taken from the residual closure of the input formula, not only from its original parse tree.

RuleRunner, fixed. Formula progression gives a principled account of what remains to be checked after reading one observation. Given φ and $s \in 2^{\mathcal{P}}$, let $\text{prog}(\varphi, s)$ denote the formula that has to hold on the remaining suffix after s has been consumed, such that for every finite continuation σ , the following holds:

$$s \cdot \sigma \models \varphi \iff \sigma \models \text{prog}(\varphi, s),$$

Formally, formula progression is defined recursively as follows:

$$\begin{aligned} \text{prog}(a, s) &= \top \text{ if } a \in s, \text{ and } \perp \text{ otherwise,} \\ \text{prog}(\varphi_1 \wedge \varphi_2, s) &= \text{prog}(\varphi_1, s) \wedge \text{prog}(\varphi_2, s), \\ \text{prog}(\varphi_1 \vee \varphi_2, s) &= \text{prog}(\varphi_1, s) \vee \text{prog}(\varphi_2, s), \\ \text{prog}(X\varphi, s) &= \varphi \wedge F\top, \\ \text{prog}(W\varphi, s) &= \varphi \vee G\perp, \\ \text{prog}(\varphi_1 U \varphi_2, s) &= \text{prog}(\varphi_2, s) \vee (\text{prog}(\varphi_1, s) \wedge (\varphi_1 U \varphi_2)). \end{aligned}$$

We assume that every time *prog* is computed, it is followed by a Boolean simplification. Derived operators are handled through their definitions.

The repaired monitor carries a single *residual formula* $\rho_t \in cl(\varphi)$, the formula that the remaining suffix must satisfy. It is initialized to $\rho_0 = \varphi$ and updated by progression

$$\rho_{t+1} = \text{nf}(\text{prog}(\rho_t, \sigma_t))$$

where nf is a semantics-preserving Boolean normalization discussed below. The verdict is read off the residual itself: $\rho_t \equiv \top$ means that every continuation satisfies φ , and the monitor halts with SATISFY; $\rho_t \equiv \perp$ means that none does, and it halts with VIOLATE; otherwise the verdict is UNDECIDED. In the eager realization, these equivalences are decided exactly from the completed residual graph. Let F be the residual states that accept the empty suffix. A state is an accepting sink precisely when no state outside F is reachable from it, and it is a trap precisely when no state in F is reachable from it. Both sets are computed once by reverse reachability. The lazy realization instead halts early only when normalization exposes the literal constants $\top$ or $\perp$; this test is sound but may return UNDECIDED longer than necessary. As in the original system, evaluating a formula at a time point is a bottom-up computation over its parse tree. In details, the RuleRunner state S_t at time t holds the observations of σ_t , the active rules $R[\psi]$, and the truth evaluations $[\psi]V$ for $\psi \in \text{sub}(\rho_t)$. These are exactly the registers the original system used, only instantiated over the residual rather than over φ ; and, exactly as before, they are *flushed* at each time point rather than carried. What is carried across time is the residual alone. Note that since only the residual is progressed, the undecided subformulas $[\psi]?$ of ρ_t are not reactivated as before, but flushed as every other truth evaluation.

The residual is carried not as a monolithic formula but in *factored* form, as the set of its top-level conjuncts:

$$\Sigma_t = \text{split}_{\wedge}(\rho_t)$$

where $\text{split}_{\wedge}(\chi_1 \wedge \chi_2) = \text{split}_{\wedge}(\chi_1) \cup \text{split}_{\wedge}(\chi_2)$, $\text{split}_{\wedge}(\top) = \emptyset$, and any formula that is not a top-level conjunction is kept as a single element. Progression is a homomorphism for conjunction, $\text{prog}(\chi_1 \wedge \chi_2, s) = \text{prog}(\chi_1, s) \wedge \text{prog}(\chi_2, s)$, so the elements of Σ_t may be progressed independently and re-collected, leaving the update above unchanged. The factorization is used to reduce the size of the system. If we carry the whole residual, the monitor needs one register per reachable residual, i.e. it has the size of $cl(\varphi)$. Let $\mathcal{R}_{\varphi} = \bigcup_{\rho \in cl(\varphi)} \text{split}_{\wedge}(\rho)$ be the residual-root vocabulary. Carrying conjuncts instead gives a *multi-hot* vector over $\mathcal{R}_{\varphi}$, and the gap can be exponential. For $\varphi = Fa_1 \wedge \dots \wedge Fa_n$, the reachable residuals are all subsets of still-pending conjuncts, so $|cl(\varphi)| = 2^n$, whereas $|\mathcal{R}_{\varphi}| = n$.

To summarize, the RuleRunner two-phase schema is repaired as follows:

1. **Evaluation.** The observation σ_t is injected into S_t and the evaluation rules R_E are fired bottom-up, computing $[\psi]V$ for the syntactic closure of the residual roots. These values provide the end-of-trace decision for the current cell; ongoing state evolution is performed by the progression-based reactivation below.
2. **Reactivation.** The residual is factorized and each active root is progressed independently. Each result is normalized and split into top-level conjuncts, and the emitted roots are unioned into the next recurrent state:

$$\Sigma_{t+1} = \bigcup_{\chi \in \Sigma_t} \text{split}_{\wedge}(\text{nf}(\text{prog}(\chi, \sigma_t))), \quad \rho_{t+1} = \bigwedge_{\eta \in \Sigma_{t+1}} \eta.$$

This update is the structured progression network's reactivation phase. A clause owned by χ tests only $R[\chi]$ and the observations relevant to χ , and emits the corresponding $R[\eta]$ registers. No transition clause recognizes the complete set Σ_t . Cross-root simplifications need not be performed for correctness: for example, $\{a, \neg a\}$ already denotes the unsatisfiable conjunction even if it is not rewritten to $\{\perp\}$. The reachable sets of roots are nevertheless enumerated at compilation time to

label traps and accepting sinks exactly; this fixed global readout determines only the online verdict and does not drive recurrence.

The cost of the repair. One may wonder whether the new RuleRunner construction comes with a cost. Actually, it depends on how the monitor is realized.

In the *eager* realization, the residual closure $cl(\varphi)$ and its syntactic closure $sub(cl(\varphi)) = \bigcup_{\chi \in cl(\varphi)} sub(\chi)$ are computed once, and all corresponding evaluation and progression rules are compiled into a fixed finite rule system. Using the CILP translation, we can then map it to a recurrent network using the same two-phase scheme as the original RuleRunner. The flat realization compiles whole-residual transitions, whereas the structured realization above compiles one transition module per residual root and combines their outputs. The price is paid on two axes. First, the recurrent register set is bounded by $|\mathcal{R}_\varphi|$, rather than by the syntactic subformulae of φ . Second, $\text{prog}(\chi, \sigma_t)$ depends on the observation, so compiling a root's transition clauses may require expanding over $2^{|\mathcal{P}_\chi|}$ valuations of the atoms relevant to that root. Exact early verdicts additionally require the reachable aggregate root sets to be classified by the fixed global readout. Both the root vocabulary and this aggregate-state graph can be exponential in the worst case. The observation expansion is analogous to the alphabet axis of the dense DeepDFA tensor (see Sec. 4).

In the *lazy* realization, only the closure $sub(\rho_t)$ of the current residual is instantiated, one time point at a time. Neither axis is ever enumerated: the set of registers is bounded by the parse tree of ρ_t , that is $sub(\rho_t)$, and the progression is computed only for that particular residual against the current observation, not expanded over $2^{|\mathcal{P}|}$. The cost here is that we cannot encode the monitor as a recurrent network, since the rule set is no longer fixed, but dynamically generated. Conversely, we can use a *sequence of distinct feedforward neural networks*, once per time step. This means that cross-trace batching is lost, since traces in a batch hold different residuals and share no weight matrix.

Soundness and Completeness. Now, we can prove that the repaired construction of RuleRunner is sound and complete.

Theorem 3. *Let φ be an LTL_f formula and let RR_φ^{prog} be the (eager) progression-based RuleRunner monitor constructed as above. Then, for every finite trace σ , the monitor returns SATISFY iff $\sigma \models \varphi$, and it returns VIOLATE iff $\sigma \not\models \varphi$. Moreover, any verdict returned before the last cell is sound for all finite continuations of the prefix read so far.*

Proof. Let $\sigma = (\sigma_0, \dots, \sigma_{T-1})$ be a finite trace. For $0 \leq t \leq T$, let Σ_t be the active root set after the prefix $\sigma_0 \cdots \sigma_{t-1}$ has been consumed (and thus before reading σ_t when $t < T$). Let

$$\rho_t = \bigwedge_{\chi \in \Sigma_t} \chi$$

be the residual it denotes. By construction,

$$\Sigma_0 = \text{split}_\wedge(\text{nf}(\varphi)),$$

and therefore ρ_0 is equivalent to φ .

We first use the defining property of progression. For every formula χ , every cell s , and every finite continuation τ , including the empty continuation ϵ ,

$$s \cdot \tau \models \chi \iff \tau \models \text{prog}(\chi, s). \quad (3)$$

This property follows by structural induction on χ . The atomic and Boolean cases are immediate from the definition of **prog**. For $X\chi$, $s \cdot \tau \models X\chi$ iff τ is non-empty and $\tau \models \chi$. Since $F\top$ holds exactly on non-empty finite traces, this is equivalent to $\tau \models \chi \wedge F\top$. Dually, $W\chi$ is true either when the suffix is empty or when χ holds at its first cell; $G\perp$ holds exactly on the empty trace, yielding $\text{prog}(W\chi, s) = \chi \vee G\perp$. For until, $s \cdot \tau \models \chi_1 U \chi_2$ iff either χ_2 holds at the current cell, or χ_1 holds at the current cell and $\chi_1 U \chi_2$ holds on the suffix; by the induction hypothesis this is equivalent to

$$\tau \models \text{prog}(\chi_2, s) \vee (\text{prog}(\chi_1, s) \wedge (\chi_1 U \chi_2)).$$

The derived operators inherit the property from their definitions.

We now prove the residual invariant:

$$\sigma_0 \cdots \sigma_{t-1} \cdot \tau \models \varphi \iff \tau \models \rho_t \quad (4)$$

for every $0 \leq t \leq T$ and every finite continuation τ .

For $t = 0$, the claim holds because ρ_0 is equivalent to φ . Assume it holds for some $t < T$. By the reactivation rule,

$$\rho_{t+1} \equiv \text{nf} \left(\bigwedge_{\chi \in \Sigma_t} \text{prog}(\chi, \sigma_t) \right).$$

Since $\rho_t = \bigwedge_{\chi \in \Sigma_t} \chi$ and progression distributes over conjunction up to logical equivalence, this is equivalent to

$$\rho_{t+1} \equiv \text{nf}(\text{prog}(\rho_t, \sigma_t)).$$

Therefore, for every continuation τ ,

$$\begin{aligned} \sigma_0 \cdots \sigma_t \cdot \tau \models \varphi &\iff \sigma_t \cdot \tau \models \rho_t \quad \text{by the induction hypothesis,} \\ &\iff \tau \models \text{prog}(\rho_t, \sigma_t) \quad \text{by Equation (3),} \\ &\iff \tau \models \rho_{t+1} \quad \text{because nf preserves semantics.} \end{aligned}$$

Thus the invariant holds at $t + 1$.

After consuming all T cells, repeated application of the invariant gives

$$\sigma \models \varphi \iff \epsilon \models \rho_T.$$

The final evaluation rules compute precisely whether the residual accepts the empty trace. Hence the monitor returns SATISFY exactly on traces satisfying φ , and VIOLATE otherwise.

Finally, the operational register construction realizes the symbolic transition used in the proof. At time t , the state $S_t = \text{sub}(\rho_t)$ contains every subformula needed to evaluate and progress the active roots. Since the eager monitor allocates registers for all formulas in $\text{sub}(cl(\varphi))$, every such support formula has a corresponding register, and the bottom-up computation can be carried out by finitely many rules. Only the active roots in Σ_t are reactivated, so support subformulas do not acquire an independent temporal meaning. Therefore the rule-based monitor implements exactly the residual update used in the invariant.

If an early verdict is returned, the current residual is respectively valid or unsatisfiable over all finite continuations. By Equation (4), every completion of the prefix read so far then has the same truth value for φ , which proves soundness of early termination. $\square$

The role of normalization. As described above, we normalize residual formulas after progression. This is not required for semantic soundness—taking nf to be the identity still preserves the progression invariant—but raw syntax may then grow without bound. For instance, repeated failure of ψ in $F\psi$ would produce $\perp \vee F\psi$, then $\perp \vee (\perp \vee F\psi)$, and so on. Normalization removes this irrelevant growth and gives the eager construction a finite structural vocabulary.

The implementation first applies smart-constructor identities that are valid also on the empty suffix: negations are pushed through the X/W , F/G , and U/R dual pairs; $FF\psi = F\psi$ and $GG\psi = G\psi$; $X\perp = \perp$ and $W\top = \top$; and the safe constant cases $\top U\psi = F\psi$, $\psi U\perp = \perp$, $\perp R\psi = G\psi$, and $\psi R\top = \top$ are folded. It then simplifies the Boolean skeleton while treating maximal temporal subformulas as opaque variables. This keeps residuals compact, exposes more top-level conjuncts to $\text{split}_\wedge$, and can expose $\top$ or $\perp$ for sound lazy early termination.

This normal form is deliberately syntactic: it is not canonical modulo full LTL_f language equivalence. Distinct residuals may therefore have the same right language, and raw closure size must not be presented as minimal-DFA size. Exact eager early termination is unaffected, since it uses the graph-theoretic sink/trap test rather than relying on syntactic exposure of $\top$ or $\perp$.

The counterexample revisited. Consider again the formula $\varphi_b = F(a \wedge Xb)$ and the two traces $\sigma_A = (\{a\}, \emptyset, \{b\})$ and $\sigma_B = (\emptyset, \{a\}, \{b\})$. The residual formulas before each time step are:

$$\begin{array}{ccccccc}
\rho_0 & & \xrightarrow{\sigma_0} \rho_1 & & \xrightarrow{\sigma_1} \rho_2 & & \xrightarrow{\sigma_2} \rho_3 \\
\sigma_A : F(a \wedge Xb) & \xrightarrow{\{a\}} (b \wedge F\top) \vee F(a \wedge Xb) & \xrightarrow{\emptyset} F(a \wedge Xb) & \xrightarrow{\{b\}} F(a \wedge Xb) \\
\sigma_B : F(a \wedge Xb) & \xrightarrow{\emptyset} F(a \wedge Xb) & \xrightarrow{\{a\}} (b \wedge F\top) \vee F(a \wedge Xb) & \xrightarrow{\{b\}} \top
\end{array}$$

For trace σ_A , the residual $(b \wedge F\top) \vee F(a \wedge Xb)$ appears after the first time step. The marker $F\top$ records that the pending b obligation requires another cell. At the second time step, b is false, so the disjunct disappears and the residual becomes $F(a \wedge Xb)$ again. After the last time step the same eventuality remains, and it is false on the empty suffix. Hence the verdict is **VIOLATE**, as expected. For trace σ_B , the residual $(b \wedge F\top) \vee F(a \wedge Xb)$ appears one step later, immediately before the time step containing b . At the last time step, b holds and $F\top$ progresses to true because that cell exists. Hence the residual becomes $\top$ and the verdict is **SATISFY**.

This is exactly the information lost by the original one-register construction. The repaired monitor does not try to store two incompatible truth evaluations in the single register $[Xb]$. Instead, it stores the residual formula $(b \wedge F\top) \vee F(a \wedge Xb)$, and it occurs at different time points in the two traces.

4 Automata-based Neuro-Symbolic Monitoring: DeepDFA

4.1 Architecture

DeepDFA [17] keeps the automaton of the symbolic paradigm but replaces its explicit state walk with a differentiable, tensorized one; we adopt the same forward formulation used for predictive monitoring of business processes by [12]. In its original formulation DeepDFA is a *probabilistic relaxation* of a DFA: a probabilistic finite automaton whose transition logits and acceptance logits are real-valued and *learnable*. Temperature-scaled softmax constrains each transition row to the

probability simplex, temperature-scaled sigmoid produces the acceptance vector, and annealing drives both toward the one-hot/Boolean limit. This is distinct from the Gumbel–Softmax estimator used by [12] to sample differentiable near-symbolic *inputs* for a fixed automaton [7, 9]. Accordingly, starting from the minimal DFA $A_\varphi = (2^{\mathcal{P}}, Q, q_0, \delta, F)$ of Section 2.2, the general object is a three-way tensor $T \in [0, 1]^{|Q| \times |2^{\mathcal{P}}| \times |Q|}$ that is row-stochastic in its last index, $\sum_{q'} T[q, \sigma, q'] = 1$, together with an acceptance vector $v_o \in [0, 1]^{|Q|}$. When the automaton is instead *compiled* from a known LTL_f formula — the setting of this paper’s monitoring experiments, and the fixed knowledge layer of [12] — these quantities are fixed to their exact one-hot values, $T \in \{0, 1\}^{|Q| \times |2^{\mathcal{P}}| \times |Q|}$ with $T[q, \sigma, q'] = 1$ iff $\delta(q, \sigma) = q'$, and $v_o \in \{0, 1\}^{|Q|}$ with $v_o[q] = 1$ iff $q \in F$. Learning them is unnecessary for monitoring a known specification; a separate real-valued parameterization would instead be needed for the specification-adaptation problem deferred to Section 8. In either case the monitor state is no longer a single automaton state but a row vector $\mathbf{q}_t \in [0, 1]^{|Q|}$ over states, initialized to the one-hot vector $\mathbf{q}_0 = v_i$ of q_0 . Writing $T_\sigma = T[\cdot, \sigma, \cdot] \in [0, 1]^{|Q| \times |Q|}$ for the (row-stochastic) transition matrix of symbol σ , a step is a single vector–matrix product

$$\mathbf{q}_t = \mathbf{q}_{t-1} T_{\sigma_t}, \quad \text{acc}_t = \mathbf{q}_t \cdot v_o, \quad (5)$$

where $\text{acc}_t \in [0, 1]$ is the acceptance score of the prefix read so far. The following elementary invariant records exactly what is preserved by the tensorization.

Proposition 1 (Crisp equivalence). *Let $\sigma_{1:t} = \sigma_1 \cdots \sigma_t$ be a crisp prefix and let $\mathbf{q}_t$ be obtained from Equation (5) using the compiled one-hot tensor. Then*

$$\mathbf{q}_t = e_{\delta^*(q_0, \sigma_{1:t})},$$

where e_q is the indicator vector of state q . Consequently the fixed DeepDFA monitor and the symbolic DFA visit the same state after every prefix and return identical early and final verdicts.

Proof. At $t = 0$, $\mathbf{q}_0 = e_{q_0}$. If $\mathbf{q}_{t-1} = e_q$, the compiled row $T[q, \sigma_t, \cdot]$ has its unique one at $\delta(q, \sigma_t)$, hence $\mathbf{q}_t = e_q T_{\sigma_t} = e_{\delta(q, \sigma_t)}$. Induction proves the invariant. Trap, accepting-sink, and final-state membership are functions of this same reached state, so the verdicts coincide. $\square$

4.2 Soft Observations and Differentiability

Beyond crisp monitoring, the tensorization admits probabilistic observations. For $\mathbf{p} \in [0, 1]^{\mathcal{P}}$ and valuation $\sigma \subseteq \mathcal{P}$, define the independent-Bernoulli weight

$$w_{\mathbf{p}}(\sigma) = \prod_{a \in \sigma} p_a \prod_{a \in \mathcal{P} \setminus \sigma} (1 - p_a). \quad (6)$$

If $g_{q,q'}$ is the disjunction of the guards on all edges from q to q' , the exact expected transition is

$$M(\mathbf{p})[q, q'] = \sum_{\sigma \models g_{q,q'}} w_{\mathbf{p}}(\sigma). \quad (7)$$

Thus a per-atom probability vector produced by a perceptor or sensor replaces one-hot symbol selection while leaving the forward update unchanged:

$$\mathbf{q}_t = \mathbf{q}_{t-1} M(\mathbf{p}_t), \quad \text{acc}_t = \mathbf{q}_t \cdot v_o. \quad (8)$$

The general construction and its state-distribution semantics are already established for symbolic automata by NeSyA [10]. We record the specialization needed here to make the assumptions and the correctness of our LTLf monitor explicit.

Proposition 2 (Exact guard WMC). *Let $\mathcal{C}_{q,q'}$ be a pairwise-disjoint cube cover of $g_{q,q'}$. For a cube c , let c^+ and c^- be its required-true and required-false atoms and set*

$$w_{\mathbf{p}}(c) = \prod_{a \in c^+} p_a \prod_{a \in c^-} (1 - p_a).$$

Then $M(\mathbf{p})[q, q'] = \sum_{c \in \mathcal{C}_{q,q'}} w_{\mathbf{p}}(c)$ equals Equation (7) for every Boolean guard. Moreover, $M(\mathbf{p})$ is row-stochastic and differentiable in $\mathbf{p}$.

Proof. Each cube denotes a cylinder set of complete valuations and its product is the total Bernoulli weight of that set. The cubes are disjoint and their union is exactly the models of $g_{q,q'}$, so their weights sum to Equation (7). For fixed q , determinism and completeness of the DFA make the outgoing guards a partition of $2^{\mathcal{P}}$; therefore $\sum_{q'} M(\mathbf{p})[q, q'] = \sum_{\sigma \subseteq \mathcal{P}} w_{\mathbf{p}}(\sigma) = 1$. Each entry is a finite multilinear polynomial in $\mathbf{p}$ and is therefore differentiable. $\square$

Proposition 3 (Trace-marginal semantics). *Let $\Sigma_1, \dots, \Sigma_L$ be independent random valuations, with $\Pr(\Sigma_t = \sigma) = w_{\mathbf{p}_t}(\sigma)$, and propagate Equation (8) with the exact matrices of Equation (7). Then, for every $q \in Q$,*

$$\mathbf{q}_t[q] = \Pr(\delta^*(q_0, \Sigma_{1:t}) = q).$$

Consequently $\mathbf{q}_L \cdot v_o = \Pr(\Sigma_{1:L} \models \varphi)$.

Proof. The claim is immediate at $t = 0$. Assuming it at $t - 1$, the law of total probability and independence of Σ_t from the preceding cells give

$$\Pr(\delta^*(q_0, \Sigma_{1:t}) = q') = \sum_{q \in Q} \mathbf{q}_{t-1}[q] M(\mathbf{p}_t)[q, q'],$$

which is exactly the q' component of Equation (8). Summing the final distribution over F , equivalently taking its inner product with v_o , yields the acceptance marginal. $\square$

Because the cube weights are smooth, the fixed monitor is differentiable with respect to its observations: a verdict loss can flow through it into an upstream perceptor without changing the specification. Making the specification itself learnable is a separate construction requiring parameterized transitions or guards, an adaptation objective, and a discrete extraction policy; we defer that problem to Section 8. In this paper all evaluated traces are crisp, where $M(\mathbf{p}_t)$ collapses to T_{σ_t} and Proposition 1 applies.

4.3 The Alphabet Blow-up

DeepDFA's structural cost is dual to those of the other two paradigms. The transition tensor is indexed by the full alphabet $2^{\mathcal{P}}$, so a dense materialization is exponential in the number of atoms, $|T| = |Q|^2 \cdot 2^{|\mathcal{P}|}$. One can avoid building the dense tensor by evaluating the per-edge guard probabilities on the fly — the factored form underlying Equations (5)–(8), which never explicitly materializes T — and this is what we use for specifications over many atoms. This

does not guarantee a sub-exponential representation for every Boolean function: a guard may require exponentially many disjoint cubes. Read-once structure is one sufficient condition for linear evaluation, but it is not necessary; other guards can admit compact orthogonal covers or decision diagrams through repeated-subfunction sharing.

Concretely, the monitor admits two representations of the transition function, and we evaluate both. The *dense* representation materializes the full tensor T and executes each step as a single (batched) matrix product indexing the observed symbol; it is the fastest option whenever the alphabet fits in memory and is the natural showcase for cross-trace batching, but it becomes infeasible past roughly a dozen atoms, where $2^{|\mathcal{P}|}$ exhausts memory. The *factored* representation, developed next, never builds T and is what we use for specifications over many atoms.

4.4 The Factored Transition Representation

The factored representation exploits the fact that the DFA edges produced by MONA are labelled not by individual symbols but by *symbolic Boolean guards* over the atoms (e.g. $a \wedge (b \vee c)$), each standing for the set of symbols that satisfy it. This is the same compact edge labelling that lets the symbolic walk sidestep the alphabet (Section 2.3); the factored mode carries it into the tensorized setting while preserving exact crisp monitoring and a differentiable soft path. NeSyA [10] already gives the more general construction: symbolic-automaton guards are compiled, e.g. to d-DNNF, and evaluated by exact differentiable WMC before the same matrix forward recursion. We neither claim that semantics nor knowledge compilation as new. Our contribution here is a concrete specialization to DFAs compiled from LTL_f runtime specifications, using disjoint cubes as a simple tensor backend and placing its costs beside dense DeepDFA, symbolic monitoring, and RuleRunner. The cube construction uses standard Shannon expansion and disjoint sum-of-products [3]; unlike NeSyA’s shared circuits, it provides no subfunction sharing.

We keep two complementary views of each edge guard: an exact disjoint representation used by the monitor, and a recursive approximation retained for analysis.

Exact path: disjoint cube cover. At construction time each guard is expanded, by Shannon expansion, into a set of *disjoint* (mutually exclusive) cubes — partial assignments in which some atoms are required true, some required false, and the rest are don’t-cares — stored as a pair of require-true/require-false integer masks over the atoms. Given a per-atom value vector $\mathbf{p}$, each cube evaluates to $\prod_a [1 - \text{rt}_a(1 - p_a) - \text{rf}_a p_a]$ (which is p_a for a required-true atom, $1 - p_a$ for a required-false atom, and 1 for a don’t-care), and the cubes of all edges are summed into the $|Q| \times |Q|$ transition matrix by a single vectorized reduction, at a cost proportional to the number of cubes rather than to $2^{|\mathcal{P}|}$. On crisp 0/1 inputs each cube is 0/1, so this path reproduces the symbolic walk exactly. On fractional inputs each cube product is its probability under the independent-atom model; because the cubes are mutually exclusive, their sum is the exact guard probability for arbitrary Boolean guards and the resulting matrix remains row-stochastic. The same tensor operations are differentiable in $\mathbf{p}$. This is the path our monitoring experiments time and the default path exposed by the tensor-native probabilistic API.

Crucially, the factored representation is *not* an unconditional escape from the alphabet blow-up: Shannon expansion of a guard on k atoms can produce up to $\Theta(2^k)$ disjoint cubes, so a guard with no compact orthogonal cover reinstates the exponential cost in the number of cubes. What the factored form buys is a shift of the blow-up from “always $2^{|\mathcal{P}|}$ ” (the dense tensor) to “structure-dependent”: a genuine and often large win on guards with compact covers, including the factored

Representation	Static transition storage	Per-cell time	Fractional input
Dense tensor	$\Theta(m^2 2^n)$	$\Theta(Bm^2)$	not exposed
Disjoint cubes	$\Theta(Cn)$	$\Theta(B(Cn + m^2))$	exact
Recursive closures	$\Theta(S)$	$\Theta(B(S + m^2))$	exact for read-once guards

Table 1: Complexity of the implemented DeepDFA transition paths. The cube implementation stores dense require-true/require-false masks of length n , hence $\Theta(Cn)$ rather than the number of mentioned literals.

read-once families used in our breadth experiments, and no asymptotic improvement on adversarial ones.

Recursive approximation. Separately, each guard is compiled into a differentiable closure that applies the familiar independence identities recursively over its Boolean syntax, including $P(\psi_1 \wedge \psi_2) = P(\psi_1)P(\psi_2)$ and $P(\psi_1 \vee \psi_2) = 1 - (1 - P(\psi_1))(1 - P(\psi_2))$. This is exact for crisp inputs and for fractional inputs when the relevant subexpressions are independent, in particular for read-once guards. On non-read-once guards, however, shared atoms correlate the subexpressions and the recursion can double-count mass. We therefore retain it only as an explicitly selected diagnostic approximation; the probabilistic monitor defaults to the exact disjoint-cube calculation above.

Implementation-level complexity. Let $n = |\mathcal{P}|$, $m = |Q|$, B be the trace-batch size, C the total number of stored disjoint cubes over all DFA edges, and S the total number of Boolean-syntax nodes in the recursive guard closures. The bounds in Table 1 describe the implemented tensor paths for one cell; they include constructing the $B \times m \times m$ transition matrices and applying them to the state batch.

These are post-compilation bounds; the cost of symbolic simplification during Shannon expansion depends on the compiler. The emitted representation alone can have $\Theta(2^n)$ cubes for one guard and $C = \Theta(m2^n)$ cubes over a complete DFA in the worst case. The cube path therefore changes an unconditional alphabet tensor into a structure-dependent compilation; it does not improve the worst case. Its per-cell temporary storage is $\Theta(B(Cn + m^2))$. The implemented dense backend selects a complete crisp valuation in $\Theta(Bm^2)$ time. An exact dense calculation from independent per-atom probabilities would instead mix all 2^n symbol matrices and cost $\Theta(Bm^2 2^n)$ per cell; this interface is not implemented because the factored path computes the same marginal without unconditional alphabet enumeration.

For a length- L batch, the sequential recurrence multiplies the corresponding per-cell time by L . For $L \geq 2$, the optional Hillis–Steele prefix scan instead stores $\Theta(LBm^2)$ matrices and uses $\Theta(\log L)$ parallel stages of matrix–matrix products, for $\Theta(BLm^3 \log L)$ total arithmetic work. It reduces sequential launch depth rather than asymptotic work, which is why we treat it as a hardware-dependent variant rather than a complexity improvement.

Artifact interface. The released monitor exposes dense, cube-factored, and prefix-scan variants through separate classes. Its tensor-native probabilistic entry point accepts either an $L \times n$ trace or a padded $B \times L \times n$ batch with explicit lengths, preserves autograd to the input probabilities, and defaults to exact cube WMC; the recursive approximation requires an explicit option. The API rejects non-finite values and values outside $[0, 1]$. Each compiled monitor also exposes a serializable

diagnostics record containing n , m , the alphabet size, transition count, cube count where applicable, and the actual tensor element and byte counts used by the selected representation. These fields make the representation-size claims independently auditable without relying on private implementation state.

Our scalability experiments plot the dense and factored paths side by side — the dense curve walling out at the $2^{|\mathcal{P}|}$ memory bound while the factored curve stays flat on the read-once families — which separates the alphabet blow-up as a *memory* wall from the residual per-step *compute* cost.

5 Theoretical Comparison

Having presented the three paradigms individually, we now compare them along the dimensions that matter for a monitor: what each can represent, and where each breaks down.

5.1 Three Achilles Heels

Each paradigm is exact and efficient on a large class of specifications, and each has a single structural weakness that is dual to the others', so that no one paradigm dominates.

- **Symbolic DFA — state-space blow-up.** The LTL_f -to-DFA translation is doubly exponential in the worst case, so the compiled automaton (and hence the compile-time cost and memory) can explode even when the runtime visits only a handful of states.
- **RuleRunner — temporal-instance conflation.** The one-register-per-subformula state can merge evaluations of the same subformula originating at different temporal offsets, so it returns wrong verdicts when that merged register is not a sufficient semantic summary (Section 3.2). The exact safe class is decidable per formula, bounded-future instances admit a fixed pipeline repair (Section 3.3), and full LTL_f completeness is recovered through the residual vocabulary of progression (Section 3.4).
- **DeepDFA — explicit transition cost.** The dense transition tensor indexes all $2^{|\mathcal{P}|}$ valuations, while our factored backend replaces that unconditional tensor by C disjoint guard cubes. This is often compact but remains exponential in the worst case, and both realizations retain the $|Q|^2$ matrix update (Section 4.3, Table 1).

Two clarifications keep this comparison honest. First, the state-space blow-up is not exclusive to the symbolic paradigm: DeepDFA is compiled from the *same* minimal DFA, so it inherits $|Q|$ (its tensor is $|Q|^2 \cdot 2^{|\mathcal{P}|}$) and suffers the same doubly-exponential worst case. We list state blow-up under the symbolic monitor because it is that monitor's *only* structural weakness, whereas for DeepDFA the *additional*, representation-specific cost that the symbolic monitor escapes is the alphabet axis. Second, the two neuro-symbolic-versus-symbolic heels are duals in a precise sense: a DFA transition function is a table with $|Q| \cdot 2^{|\mathcal{P}|}$ entries, and each paradigm compresses a different axis of it to obtain its advantage, paying on the axis it leaves explicit. The symbolic monitor compresses the $2^{\mathcal{P}}$ axis with symbolic guards — keeping an explicit current state, so its cost driver is $|Q|$ — and this is exactly why it cannot suffer an alphabet blow-up: it never enumerates symbols. DeepDFA (dense) compresses the state axis into a uniform, differentiable, batchable matrix product — which requires one explicit transition matrix per symbol, so its cost driver is $2^{|\mathcal{P}|}$. Neither weakness can be

swapped onto the other paradigm without discarding the very representational choice that defines it; the factored representation (Section 4.4) attempts to compress *both* axes at once, and recovers flat-in- $|\mathcal{P}|$ scaling whenever the guards admit compact disjoint covers. Read-once structure is one sufficient case, not a necessary boundary.

One further symmetry is worth stating explicitly, because it is easy to present as an advantage of one paradigm. Exact *three-valued timing* — announcing SATISFY or VIOLATE at the first prefix after which the verdict is permanent — is a property of the continuation language, not of any local state. Every paradigm therefore pays for it with a global reachability analysis: the symbolic monitor and DeepDFA precompute traps and accepting sinks on the compiled DFA, the eager progression monitor classifies its reachable residual graph, and the bounded-event monitor classifies its reachable composite states. Each of these is a fixed label head, computed once and read out per cell; none of them drives the recurrence. What distinguishes the paradigms is the size of the graph that must be classified, not whether such a graph is needed at all, and a monitor that skips the analysis pays in latency rather than in soundness.

As monitors of crisp traces, the completeness-restored RuleRunner and fixed DeepDFA are semantically interchangeable: the progression construction and the minimal DFA recognize the same LTL_f language (Section 3.4), although the reachable residual graph need not itself be minimal. The reason to keep both lineages is what each could expose beyond crisp monitoring: a future learned DeepDFA would parameterize transitions between opaque automaton states, while a future rule-based learner could attach parameters to syntactically-indexed rules. These are two hypotheses for the gradient-based specification adaptation deferred to future work (Section 8), not capabilities evaluated here.

TODO: capability matrix summarizing exactness, soft-input support, differentiability, and the three heels across the three paradigms.

TODO: cite Bacchus–Kabanja progression and the LTL_f 2EXP bound as primary sources.

6 Experimental Comparison

All three monitoring paradigms are implemented in a common framework sharing a uniform MONITOR interface, which ensures that the experimental harness is paradigm-agnostic: timing experiments iterate over a list of monitor instances without any paradigm-specific branching. The shared compilation pipeline uses `l1f2dfa` [6] to translate each LTL_f formula into a minimal DFA once at construction time.

We evaluate the monitoring paradigms along three axes, using synthetic traces throughout Experiments 1–3 to isolate per-step monitoring cost from data-dependent effects such as early-termination frequency. This follows the methodology of experiments in [14].

7 Related Work

Finkbeiner and Kuhtz separate LTL specifications into maximal bounded-future subformulas and an unbounded prefix, evaluate the bounded part through a fixed pipeline, and compose both parts into a monitor circuit [5]. Our bounded-event RuleRunner adopts this temporal horizon and separation principle rather than claiming the pipeline itself as a new construction. The difference lies in the prefix component and the guarantee used to compose it: we retain the original RuleRunner architecture only when an exact product check certifies its language equivalence and prefix soundness,

and otherwise fall back to the complete progression repair.

Differentiable temporal knowledge. DeepDFA learns a probabilistic relaxation of a finite automaton by placing temperature-scaled softmax distributions on transitions and sigmoid outputs on acceptance, then annealing toward a discrete machine [17]. The same forward algebra can instead encode a known automaton as fixed one-hot parameters. NeSy predictive process monitoring uses this fixed form as a differentiable knowledge evaluator for generated suffixes [12]; our DeepDFA baseline uses it for online monitoring of a known LTL_f specification and does not implement automaton induction.

NeSyA generalizes this probabilistic forward recursion to symbolic automata whose edges carry Boolean guards [10]. It evaluates guard probabilities by exact weighted model counting over compiled representations such as d-DNNF, proves that the propagated state mass equals the marginal mass of the corresponding symbolic traces, and uses the result to train perceptual models. Our expected-transition and trace-marginal propositions are direct specializations of that semantics. The difference is the task and the concrete backend: we start from an LTL_f -compiled runtime monitor, use disjoint cubes without circuit sharing, and evaluate its crisp cost against symbolic and rule-based monitors. We therefore claim neither exact probabilistic symbolic-automaton inference nor knowledge compilation as new.

T-ILR takes the complementary formula-level route [1]. It evaluates LTL_f directly over fuzzy traces with Zadeh semantics and applies iterative local refinement, avoiding construction of an external DFA. Its output is a fuzzy satisfaction/refinement signal rather than the acceptance marginal of a probabilistic automaton. This semantic distinction is central to our comparison: in the present paper we evaluate exact crisp runtime verdicts, while probabilistic and fuzzy-input learning are separate capabilities rather than interchangeable monitor semantics.

8 Conclusion and Future Work

In this paper ...

8.1 Transformers

8.2 Specification Adaptation

8.3 Process Model Repair

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